Online Food Delivery Analysis: Data-Driven Business Insights

Customer & Order Analysis

13.Peak hour demand analysis

The peak-hour flag marks about 50% of orders as peak and shows no relationship to day, city, cuisine, order value, cancellations, delivery time or profit. Peak and off-peak orders are almost identical. About half of orders (49.9%) are marked as peak.

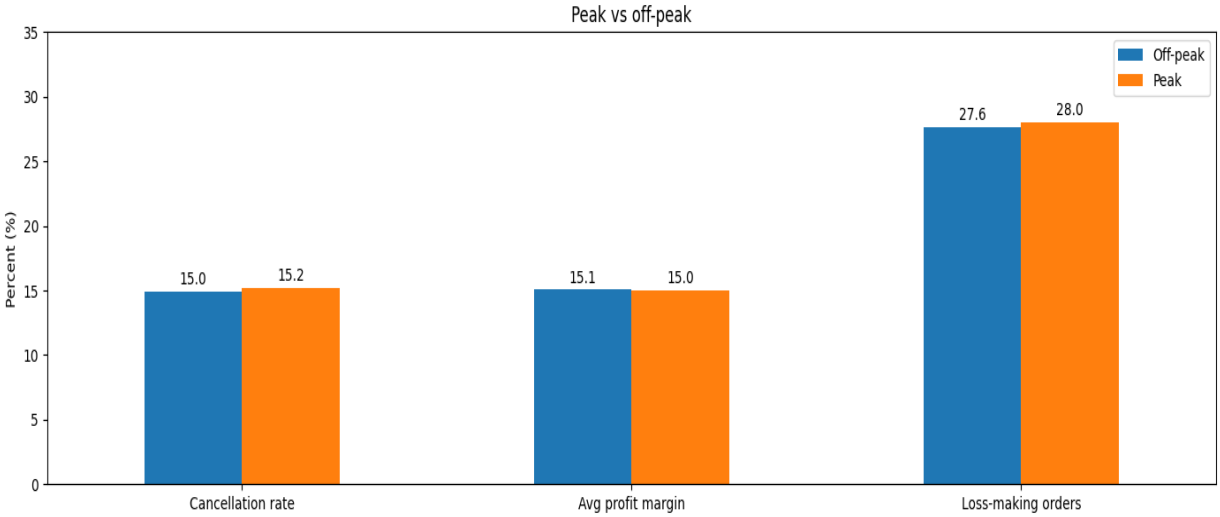
Illustrative sample records:

Peak	orders	Av order value	Cancel rate	Profit margin	Delivery time
False	33585	2082.57	14.95	15.06	128.09
True	33453	2081.38	15.19	15.01	126.76

2 . VISUALIZATIONS

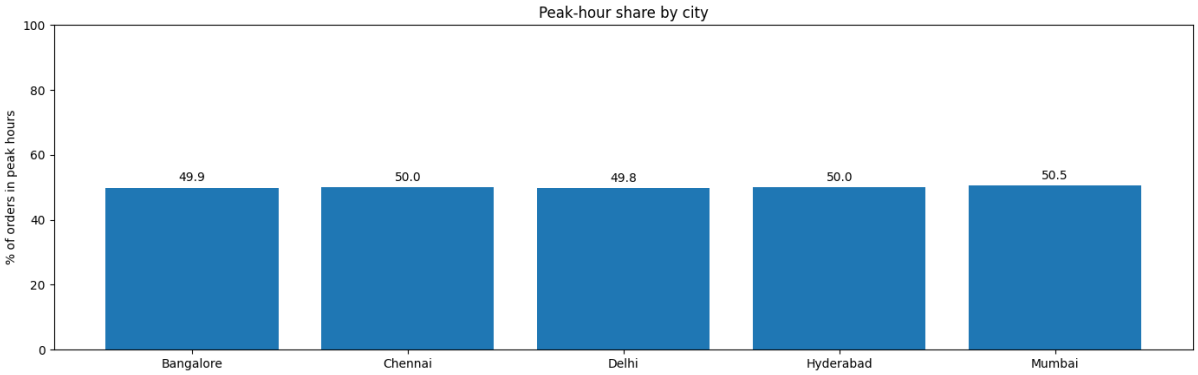
Visualization – 1

The bar chart shows the same thing for three measures: the cancellation rate, the average profit margin and the share of loss-making.



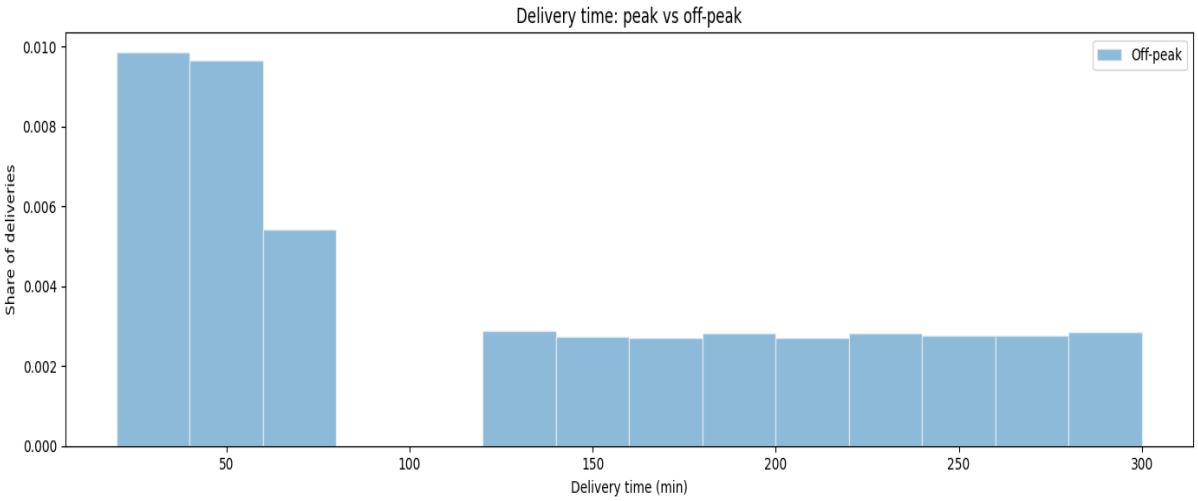
Visualization – 2

Below bar chart shows Share of orders in peak hours, by city (%).



Visualization – 3

This chart shows peak and off peak delivery time.



3.ANALYTICAL INSIGHTS – DEMAND AND REVENUE

INSIGHT – 1: Half of orders are peak.

50% of orders are marked as peak and 50%as off-peak. Demand isn't concentrated in a small busy window, as it normally would be at lunch and dinner.

INSIGHT – 2: Customers don't have a peak or off-peak habit.

A customer's share of peak orders varies, almost exactly what chance gives and only 1.5% of customers order exclusively in peak or off-peak hours. It doesn't depend on age group or payment mode either.

INSIGHT – 3: Cancellations are a bit higher at peak, but not significantly.

The rate is 15.19% at peak against 14.95% off-peak , and the lost sales are ₹4.49M against ₹4.31M. A gap of ₹0.18M is too small to justify any change.

4.ANALYTICAL INSIGHTS-OPERATIONS AND CUSTOMER EXPERIENCE:

INSIGHT-4 : Analyse Peak Hour Order Volume

Allocate more delivery partners and operational staff during high-demand hours. Increase delivery speed and quality check too.

INSIGHT- 5: Calculate Peak vs. Non-Peak Hour Revenue

Introduce targeted promotions during low-demand hours while maintaining sufficient capacity during peak periods. Increase revenue based on the peak and off peak hours.

INSIGHT- 6: Analyse Customer Ratings During Peak Hours

Identify service issues during busy periods and prioritize improvements in delivery accuracy, packaging, and communication. Compare customer ratings and negative feedback across different time slots.